

Hds Unit Manual

Page 1: Hydrodesulphurization Reactions and Catalysts

The Hydrodesulphurization (HDS) unit is a catalytic process used to remove sulfur from petroleum products (naphtha, kerosene, diesel) to comply with environmental regulations (BS-VI diesel contains less than 10 ppm sulfur). The feed is mixed with hydrogen gas, preheated, and passed through a fixed-bed reactor.

The reactor contains Cobalt-Molybdenum (CoMo) or Nickel-Molybdenum (NiMo) catalyst on alumina support. Operating conditions range from 320-380 deg C and 30-70 barg pressure. Sulfur compounds react with hydrogen to form hydrogen sulfide (H_2S), which is separated from the liquid product in a high-pressure stripper.